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Revolutionizing Traditional Oil and Gas Petrotechnical Experts Programme to Achieve Net Zero Ambitions

Z. Yew, J. Alvarez Martinez, E. Gonzalez Vega, S. Mukherjee, and M. Al-Jubouri, SLB, Abu Dhabi, United Arab Emirates; M. Ruiz, SLB, Perth, Australia; R. Alsameen, SLB, Dhahran, Kingdom of Saudi Arabia; V. De Gennaro, SLB, Pau, France; C. Chrysovilou, SLB, London, United Kingdom

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Abstract

The energy industry is grappling with a pressing trilemma: ensuring energy security, maintaining affordability, and achieving sustainability in an increasingly complex and fast-evolving landscape. Geoscientists and petroleum engineers, with their deep expertise in subsurface processes, hold significant yet underutilized potential to support emerging energy solutions. Unlocking this potential requires a fundamental shift in training approaches.

Traditional corporate learning program, built around linear e-learning, instructor-led courses, and on-the-job mentoring, has served the industry well in the past. However, the demands of the energy transition necessitate a more agile, integrated, and forward-looking training model. This paper introduces a transformative redesign of the traditional oil and gas technical curriculum into a modular, skills-based framework tailored to both traditional petroleum and new energy domains.

By systematically identifying transferable competencies—from subsurface characterization to well design and surface systems integration—a revamped program was developed to enable petrotechnical experts to support emerging applications such as carbon capture and storage (CCS), hydrogen storage, geothermal energy, and offshore renewables. The program combines classroom-based theory with practical training in industry-grade laboratories, immersive geological fieldwork enhanced by digital tools, and direct interaction with advanced acquisition and monitoring technologies.

Implemented across learning centers in Europe, the Middle East, and Asia, the program has demonstrated measurable business impact. By offering modularized pathways and fostering interdisciplinary thinking, the curriculum supports personalized learning journeys, sustained engagement, and accelerated readiness to lead in complex field development and energy transition projects.

This paper presents a proven framework for redefining corporate training to develop future-ready petrotechnical talent. It emphasizes cross-domain integration, technology adoption, and holistic skills development—setting a new benchmark for geoscience and petroleum engineering education in the era of low-carbon energy.

Introduction

The global energy sector is undergoing profound changes. As climate change intensifies and global commitments to reduce greenhouse gas emissions strengthen, countries and corporations are increasingly striving to meet ambitious net-zero targets. Simultaneously, the demand for energy continues to rise, creating a complex challenge: how to ensure energy remains affordable, secure, and sustainable.

The global push for clean energy has revealed an acute skills gap across industries. Governments and companies worldwide report a shortage of workers with "green skills," which combine technical knowledge, sustainability awareness, and digital competency needed for low-carbon operations. For instance, a recent analysis found that from 2022 to 2023, the demand for green skills in job postings grew by 22.4% ([World Economic Forum, 2024](#)). The World Economic Forum has described this mismatch between demand and supply as a "green skills crisis," driven by insufficient investment in training. Meeting global climate goals will require a massive expansion of clean-energy jobs, with the International Energy Agency (IEA) projecting over 30 million new jobs in low-carbon energy by 2030 ([IEA, 2024](#)). This underscores the urgent need to rapidly upskill the workforce.

The oil and gas sector is acutely affected by these challenges. Professionals in this field, particularly geoscientists and petroleum engineers, have long served as the technical backbone of the petroleum exploration and production lifecycle. These individuals possess deep domain knowledge of subsurface-to-surface systems, extensive field experience, and strong problem-solving capabilities grounded in physics and engineering. Their expertise has been instrumental in developing complex oil and gas reservoirs and optimizing production from both conventional and unconventional resources.

Building on their longstanding technical and ethical commitments, engineers are now expected to take a more active role in addressing environmental and social challenges through their technical decisions. In response, the organization has introduced a dedicated sustainability curriculum tailored to equip future professionals with the mindset, awareness, and foundational knowledge required to align their work with corporate climate strategies and net-zero ambitions. This training bridges engineering fundamentals with topics such as emissions awareness, digital sustainability platforms, and transition technologies. Embedding this perspective into technical training reflects a broader global imperative. For example, [UNESCO \(2022\)](#) emphasizes that integrating sustainability into engineering and vocational programs is essential to prepare the workforce for leading roles in sustainable development and climate action. This foundation sets the stage for deeper engagement with the transition technologies that make sustainable operations technically and commercially viable.

This foundational knowledge positions professionals to engage with real-world low-carbon energy technologies such as carbon capture and storage (CCUS), hydrogen storage, geothermal energy, and offshore wind, which are central to achieving near-term decarbonization goals. These technologies are reshaping the landscape of technical practice, requiring engineers and scientists to expand their competencies and adopt new ways of thinking. Applications for energy transition, particularly CCS, hydrogen, and geothermal, offer a crucial opportunity to revisit and revise traditional oil and gas methodologies and training programs. Among these, subsurface modeling for CCS is especially critical. The deployment of CCS at scale will not be feasible without rigorous modeling at all spatial and temporal scales, covering fields, wells, short-term injection behavior, and long-term storage integrity. Key decisions such as final investment decision (FID), design, execution, modeling, monitoring, measurement, and verification (MMM) rely on the comprehensive integration of physics-based models and measurements ([Sorgi, 2024](#)). These challenges are not merely about siloed disciplines such as geology, geophysics, petrophysics, geomechanics, or reservoir engineering but require individuals to possess multidisciplinary skill sets to make science-based decisions that drive project success.

To truly accelerate progress, the industry must integrate decades of research and knowledge into holistic, efficient, and rigorous solutions to develop the next generation of talent. This is essential to tackle the

urgent climate and energy challenges that impact current and future generations. The tools to enable this transformation already exist. Digital technologies, high-performance computing (HPC), and artificial intelligence now offer tangible capabilities that can validate previously overlooked solutions. In many cases, such solutions were dismissed not because of their complexity but due to technical unawareness or institutional inertia within the oil and gas sector.

Recognizing these imperatives, the organization has taken deliberate action to modernize and future-proof its corporate petrotechnical training program. The traditional curriculum has been thoroughly restructured into a modular, skills-based, and digitally enabled framework that emphasizes cross-domain integration and practical applications. This transformation is not merely academic but executed with strong business alignment, equipping the workforce with the critical skills needed to address the technical challenges of the energy transition. By combining foundational oil and gas expertise with emerging competencies in low-carbon technologies, the program ensures that employees are not only relevant but instrumental in driving the organization's journey toward a net-zero future.

Training Programme Transformation: Methodology, Structure, and Delivery

The transformation of the petrotechnical training programme followed a five-stage methodology to modernize content, integrate domains, and equip professionals for the evolving needs of the energy transition. This framework is built around three pillars: a competency-driven methodology, a modular programme structure, and blended learning modalities to enhance accessibility and engagement.

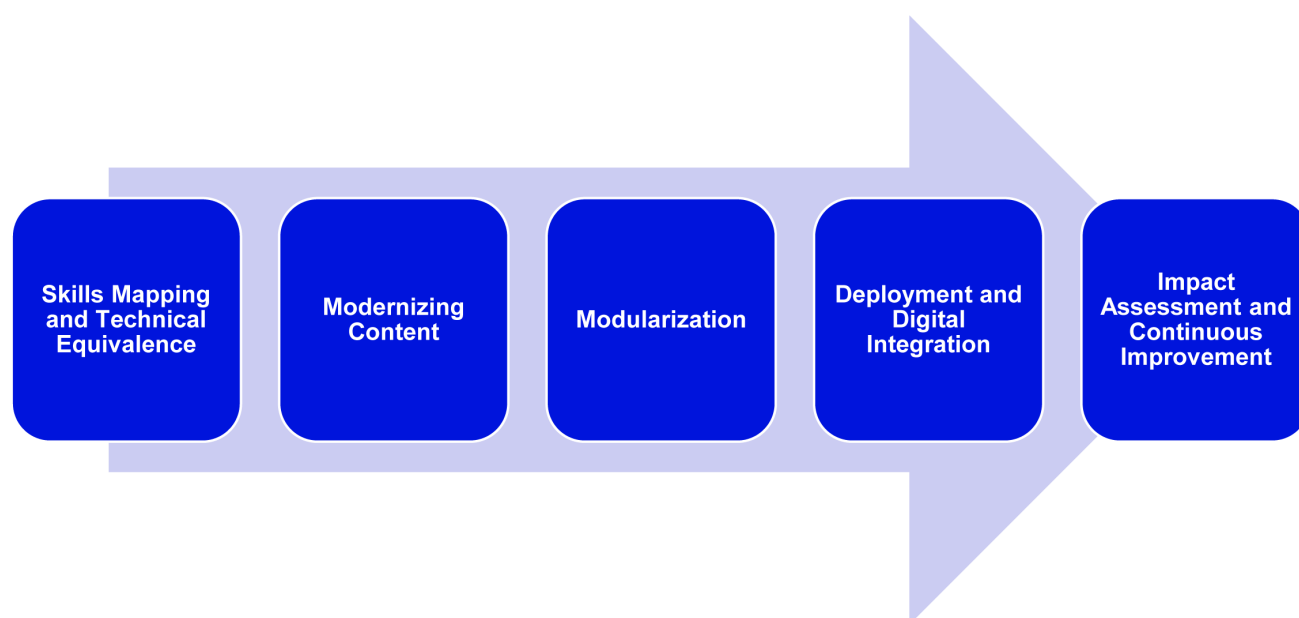


Figure 1—Methodology of petrotechnical training programme improvement

Skills Mapping and Technical Equivalence

The foundation of the transformation was a comprehensive mapping of traditional oil and gas technical skills to emerging demands in carbon capture and storage (CCS), hydrogen, offshore wind and geothermal energy. This process identified transferrable skill sets and highlighted the continuity of fundamental physics applicable to the new energy sector. The key areas of technical domains span from subsurface characterization to surface design, evaluation, and operation.

Subsurface characterization.

- i. **Seismic Interpretation** remains foundational in both oil and gas and new energy sectors, providing a subsurface image ready to be interpreted and modeled.
- ii. **Structural Modeling** which includes the interpretation and modeling of faults, horizons, and fractures, directly supports reservoir definition in traditional exploration as well as the design of containment structures in CCS and geothermal projects.
- iii. **Seismic Inversion and Rock Physics** techniques are applied to calculate and estimate elastic and velocity properties from seismic data, aiding both reservoir quality assessment and injectivity evaluation in CCS and geothermal reservoirs, being applied to offshore wind energy as well.
- iv. **Facies Modeling** is leveraged to define lithological variations that influence both flow behavior and long-term storage integrity.
- v. **Data Resampling**, such as transforming seismic cube data into simulation-ready grids and deriving well log transforms at well scale, are crucial for building coherent geocellular models.
- vi. **Properties Modeling**, including porosity, permeability, saturation and stress fields, ensures robust simulation frameworks that support storage and production forecasting.
- vii. **Coupled Thermo-Hydro-Mechanical-Chemical (THMC) Modeling** is increasingly important in modeling storage complexes and well integrity, capturing the full interaction between subsurface processes.
- viii. **Measurement, Monitoring, Modeling, and Verification (MMM)** is essential to be embedded throughout project workflows, ensuring that modeling outputs are aligned with operational realities and regulatory requirements.

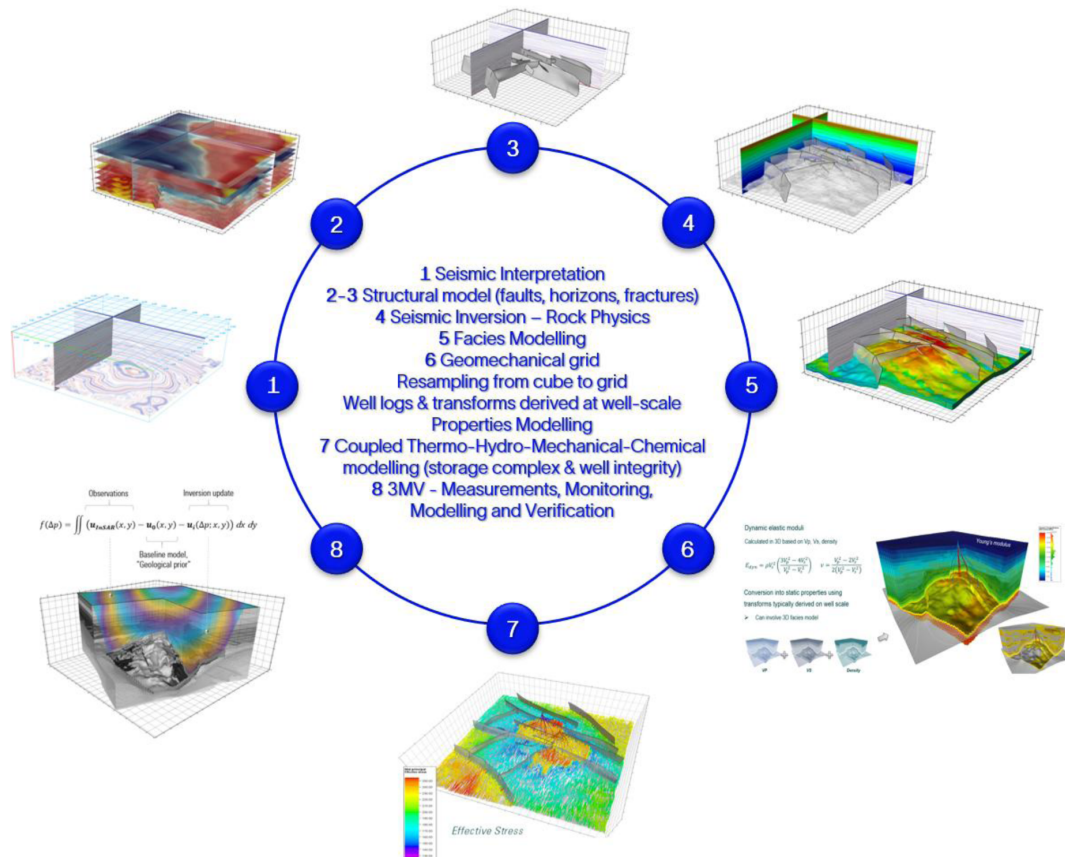


Figure 2—Subsurface characterization of integrated workflow.

Surface design, evaluation and operation.

- i. **Thermodynamics** is essential in evaluating different design alternatives of hydrogen production to improve operating potential, conversation and power demand. Similarly, this is required for assessment of carbon capture to evaluate various scenarios of carbon capture alternatives from power stations and/or industrial facilities considering CO₂ concentration, presence of impurities, low pressures and evaluating capture efficiency and power required.
- ii. **Production and processing equipment** knowledge is important to understand equipment performance. In terms of carbon capture, modeling of different equipment through specialized process engineering technologies helps to design infrastructure applied in CCS projects such as membranes, solvent absorption and absorption through solid sorbents. Furthermore, process equipment knowledge is crucial in evaluating compressor behavior to handle carbon dioxide and hydrogen to ensure energy effectiveness, operational performance considering the presence of impurities, and re-compression requirements to maintain required pressure.
- iii. **Transport through pipeline and injection:**
 - a. **Design and operational procedures:** evaluation of pipeline design to ensure safe and effective pipeline operation during normal operation, shutdown and start-ups, pigging, applying cutting-edge simulation technologies for carbon dioxide (CO₂) and hydrogen transport.
 - b. **Flow Assurance:** it is essential to apply flow assurance fundamentals to evaluate the impact of fluid composition, presence of impurities, phase behavior and flow conditions, to provide input to material and equipment selection considerations and potential risks associated with corrosion, solids formations (i.e., solid CO₂ & hydrates).
- iv. **Integrated operations through digital solutions:** application of fundamentals of flow and phase behavior, leading process engineering technologies to evaluate system performance from generation to distribution, enabling validation of control philosophy, analyzing worst case scenarios and proper mitigation plan, whilst ensuring operational safety related to pipeline integrity, risks imposed by impurities present in hydrogen and carbon dioxide.
 - a. **Digital Twin:** real-time evaluation of system performance through the application of physics-based applications and/or artificial intelligence (AI) with instrumentation that provides robust solutions for flow integrity. i.e., leak identification, phase change monitoring and investigation of pigging and line packing scenarios.
 - b. **Edge AI and IoT solutions:** deployment of digital solutions that transform field operations by bringing computing power closer to where data is generated enabling advanced real-time closed loop solutions that bridges the gap to more sustainable and safer operations, lowering the carbon and human footprint while increasing operational efficiency.

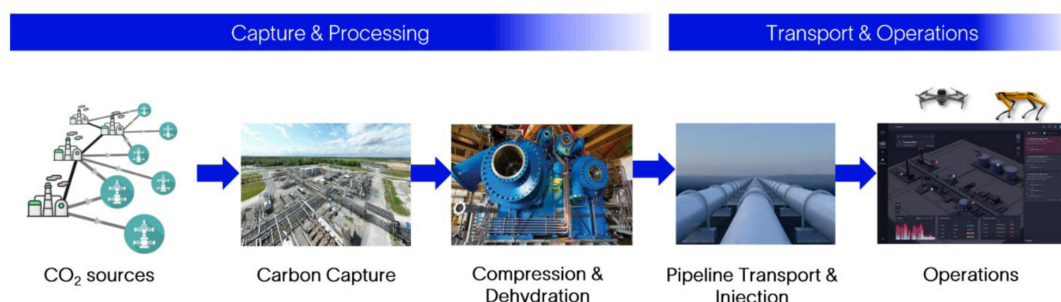


Figure 3—CO₂ surface capture, transport and injection systems end-to-end workflow

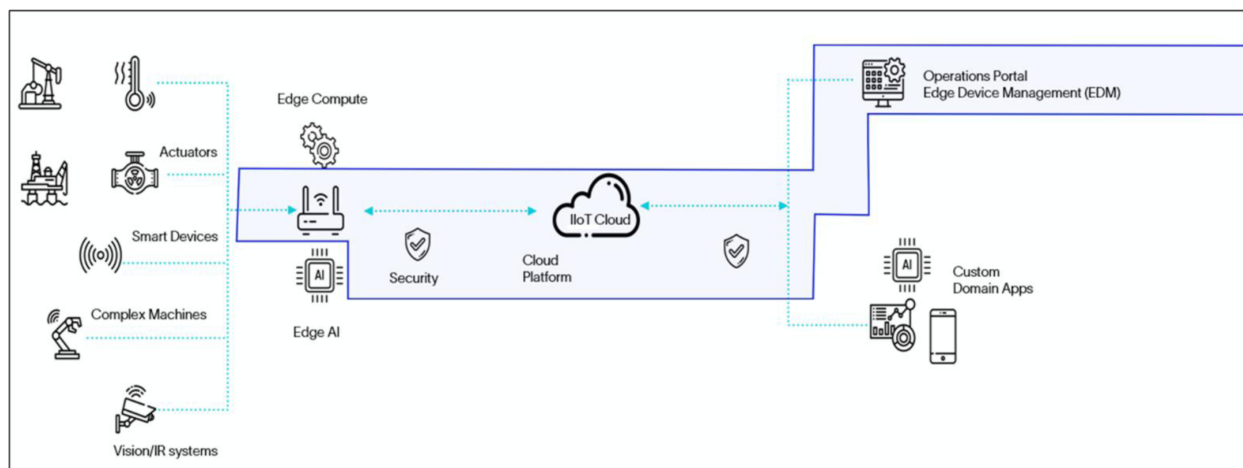


Figure 4—EDGE AI Solution ecosystem integrating the underlying elements from data acquisition to application

These skills mapping frameworks form the foundation of the revamped curriculum, reinforcing the technical continuity between oil and gas and the new energy sectors, while building the capacity for professionals to adapt and innovate. This mapping demonstrated that foundational physics and technical workflows remain relevant, though adapted in context and boundary conditions.

Modernizing Content through Domain Integration

To prepare the next generation of energy professionals for the challenges and opportunities of a low-carbon future, a new curriculum has been introduced under the Digital Sustainability modules. These modules are designed to instill a strong understanding of the science and philosophy underpinning the importance of sustainable operations and digital tools. Educating our workforce to be ready for future challenges in achieving carbon net zero goals aligns with the organization's broader corporate sustainability strategy. The goal is to position technical talent not only as skilled practitioners but also as informed advocates for sustainable development and positive environmental impact. The curriculum cultivates a mindset of technical curiosity, empowering engineers to propose sustainable solutions through the integration of digital tools and domain-specific knowledge. This includes the integration of dedicated sustainability and transition technology content addressing key themes such as community preservation, planetary stewardship, decarbonization strategies, sustainability KPOs and verified targets, as well as the enabling platforms, digital tools, and global regulatory and market outlooks that support them. Hands-on application is emphasized through digital sustainability platforms, allowing participants to practice carbon storage site screening and ranking using publicly available datasets. This ensures that learners not only understand the theory but can apply it using tools aligned with current industry standards and workflows, forming the foundation for the curriculum's broader sustainability plan.

As part of this sustainability transformation, a core component of the curriculum is developing fluency in the company's Transition Technologies portfolio, which provides innovative solutions for helping clients reduce emissions across the energy value chain. In doing so, participants gain a deeper understanding of how climate action intersects with business strategy and technical execution. In addition, the modules establish foundational knowledge in a range of decarbonization and new energy solutions, including Carbon Capture, Utilization, and Storage (CCUS), geothermal energy, offshore wind, hydrogen, and energy storage. These are directly linked to industry needs and regulatory drivers and are presented in a way that allows engineers and geoscientists to see clear pathways for applying their existing domain expertise in new contexts.

As foundational concepts are established, the curriculum advances toward the application of complex subsurface modeling techniques and surface operations, which are critical for carbon sequestration, hydrogen storage, and geothermal energy. The training adopts a modern, physics-based approach to

subsurface problem-solving, moving beyond traditional disciplinary silos like "reservoir engineering", "geomechanics" or "production engineering". Instead, it fosters a system-based mindset that equips professionals with integrated skills across geoscience and engineering domains. Subsurface engineering participants learn to construct 3D structural models, interpret petrophysical and geomechanical logs, and develop coupled simulation models that account for thermal, hydraulic, mechanical, and chemical (THMC) interactions. These integrated models reflect the true complexity of subsurface rock–fluid interactions under dynamic conditions, enabling more accurate forecasting and risk assessment in new energy projects. Whilst surface engineering participants learn to apply phase behavior and flow assurance concepts to CO₂ and hydrogen dominated systems, understanding the challenges present in its operation and application of digital technologies to improve the overall response time, increasing safety in the operation and optimize resources. By breaking down historical domain boundaries and promoting a unified, physics-grounded understanding of the subsurface, the programme cultivates a new generation of subsurface experts capable of delivering rigorous, innovative solutions to the multifaceted challenges of the energy transition.

Modular Programme Structure

In the third stage of the programme evolution, the curriculum was redesigned into modular and scalable learning units. This modularization ensures long-term sustainability and flexibility while preserving business continuity. By layering new competencies on top of the existing petroleum engineering framework, the approach retains over 80% of core technical competencies still essential to oil and gas extraction, while incorporating targeted training to address energy transition needs. This method delivers cost efficiency and minimizes disruption during workforce transformation.

Each module is self-contained, with defined learning outcomes, practical laboratory sessions, and real-world field applications. The structure enables broad-based upskilling without redundancy and supports a seamless progression from foundational to advanced expertise.

The curriculum is organized into two main tiers:

Foundation Modules. Foundation modules focus on strengthening fundamental geoscience and petroleum engineering skills that are transferable across both conventional and low-carbon energy sectors, including oil and gas, carbon capture and storage (CCS), hydrogen storage, and geothermal development. Core subjects include seismic interpretation, petrophysical log analysis, fluid and rock properties, and reservoir simulation. The integrated approach fosters both technical proficiency and strategic awareness, enabling participants to understand their role in advancing low-carbon energy goals. Hands-on training is reinforced in industry-standard labs, covering core analysis (routine and SCAL), petrology, PVT analysis, digital rock workflows, and geomechanical testing—skills crucial for reducing uncertainty in subsurface modeling. Sustainability topics are embedded early to provide context for technical learning and align trainees with the evolving energy landscape.

Examples of Foundation Modules:

- Geology and Geophysics Overview
- Petrophysical Log Acquisition and Analysis
- Reservoir Engineering Fundamentals
- Fluid Analysis with PVT Characterization
- Rock Analysis with Routine, SCAL, Geomechanics
- Reservoir Simulation Fundamentals
- Petroleum Geomechanics Fundamentals
- Flow Assurance Fundamentals

- Process Engineering Fundamentals
- Sustainability Foundation

Advanced Modules. The advanced modules emphasize multidisciplinary integration and the application of coupled modeling approaches for energy transition-related subsurface challenges. Participants are trained to unify structural modeling, petrophysical interpretation, and fluid–rock interaction processes in simulation-driven workflows.

Key focus areas include CO₂ injection and storage, hydrogen containment, and geothermal optimization, all of which require simulation of coupled thermal, hydraulic, mechanical, and chemical (THMC) interactions. Learners acquire proficiency in using commercial subsurface modeling platforms validated for CO₂ storage, enabling them to analyze geomechanical impacts such as fault stability and caprock integrity—critical for long-term containment assurance.

Production and process engineers are also trained in phase behavior and flow assurance for CO₂ and hydrogen systems. They develop expertise in simulating and optimizing the production, capture, treatment, and transport of low-carbon fluids using physics-based simulators. These tools provide real-time access to operational and engineering data across the production value chain, driving improvements in asset reliability, energy efficiency, and system performance.

The programme culminates in collaborative, project-based simulations that mirror real-world scenarios. Multidisciplinary teams address complex subsurface development challenges, fostering internal collaboration, cross-domain integration, and shared problem-solving.

Examples of Advanced Modules:

- Seismic Interpretation Assisted by Machine Learning
- Structural Modeling and Mechanical Earth Model Building
- Subsurface Property Modeling by Geostatistics and Machine Learning
- Carbon Capture and Underground Storage – Interpretation and Modeling
- Log-based CO₂ Monitoring with Pulsed Neutron Measurement
- Advanced Reservoir Simulation (Numerical Methods and HPC)
- Uncertainty Analysis and Model Selection for Production and Storage Performance Forecasting
- Digital Solutions for Energy Transition
- Unified Digital Platforms for Facilities and Infrastructure
- Coupled Modeling for Subsurface Storage (Thermal, Hydraulic, Mechanical)
- Carbon Transport and Injection Facility Design
- Hydrogen and Carbon Capture Process Design and Optimization

By preserving core oil and gas expertise while embedding new capabilities for energy transition, the modular structure equips professionals to support the industry's challenge.

Immersive and Blended Delivery

The fourth stage focused on implementing a blended delivery model that combines self-paced digital learning with immersive, instructor-led classroom training. This approach ensures scalable global reach while maintaining high engagement and hands-on application.

The programme begins with self-paced learning modules hosted on a digital upskilling platform. Learners are required to complete foundational modules in geoscience, geomechanics, reservoir engineering, and production engineering through interactive content, computer-based assessments, and peer reviews by subject matter experts. This ensures a common technical baseline before progressing to more complex topics.

Upon completion of the foundational modules, learners attend instructor-led training at global learning centers. These classroom-based sessions focus on solving real-world challenges, encouraging cross-disciplinary collaboration and innovation. Participants engage in hands-on exercises using operational tools such as real-time monitoring systems, downhole sensors, and rock and fluid laboratories.

Training culminates in immersive, multidisciplinary workshops, where attendees are tasked with delivering a comprehensive subsurface storage development plan. These workshops simulate field-scale project environments and require learners to address key technical and financial uncertainties—developing practical, team-based solutions for CCS, hydrogen storage, and geothermal energy projects. This format reinforces the application of theory to practice, preparing participants for high-impact decision-making in energy transition projects.

Impact Assessment and Continuous Improvement

To ensure that training outcomes translate into measurable workplace performance, the program incorporates a comprehensive, multi-level evaluation strategy. For the self-paced digital modules, participants follow a structured curriculum hosted on an integrated learning and upskilling platform. Their progress is systematically tracked, with a combination of online assessments and real-world application tasks reviewed by domain experts. These checkpoints serve not only to verify knowledge retention but also to confirm the learner's ability to apply concepts effectively before progressing to more advanced topics.

In the instructor-led components, assessment extends beyond technical competencies. Engineers are evaluated on several performance dimensions by a panel of seasoned subject matter experts. These include depth of technical insight, quality of deliverables, collaborative effectiveness, problem-solving agility, and the ability to innovate under realistic project constraints. This multi-dimensional evaluation framework cultivates a performance-oriented learning culture, equipping participants to function confidently in diverse and high-stakes operational settings.

Instructors also play an instrumental role in learner development by delivering timely, individualized feedback. Rather than simply scoring performance, feedback is designed to guide growth—highlighting strengths, identifying skill gaps, and encouraging reflective technical thinking. Development of conversations may include personalized coaching, one-on-one review sessions, and career-aligned planning. This ensures that learners not only stay motivated but also continuously refine their capabilities in alignment with evolving personal and organizational goals. The feedback loop supports sustained upskill and prepares engineers to adapt dynamically to the complex demands of the energy transition.

New Training Coursework and Curriculum

Sustainability Foundation

Sustainability topics are embedded early to provide context for technical learning and align trainees with the evolving energy landscape. These modules establish the foundational importance of sustainability, emphasizing alignment with global frameworks such as the United Nations Sustainable Development Goals and net-zero targets. Content explores the environmental impact of oil and gas operations and highlights the company's active strategies to mitigate climate change, conserve natural resources, and support communities in operational regions. The curriculum further introduces key sustainability metrics, digital platforms for measurement and accountability, and transition technologies that enable scalable, low-carbon innovation. The module concludes with a forward-looking view on global market trends, regulatory

shifts, and opportunities for sustainable innovation. Detailed modules covered under the Sustainability Foundation:

- i. **Introduction to Sustainability and Corporate Strategy** module covers the foundational principles of sustainability and how they align with the company's long-term corporate goals and strategic direction.
- ii. **People: Human Rights and Community Engagement** module explores the company's commitment to human rights and community development through outreach programs that promote sustainable practices in local operating environments.
- iii. **Planet: Emissions, Targets, and Environmental Stewardship** module examines the company's Scope 1, 2, and 3 emissions, climate mitigation strategies, and actions to preserve natural resources and reduce the environmental footprint of operations.
- iv. **Sustainability Metrics, KPOs, and Digital Platforms** modules introduce the company's sustainability Key Performance Objectives, measurement frameworks, and digital platforms that track progress and support decision-making.
- v. **Transition Technologies for Emissions Reduction** module highlights the role of transition technologies such as CCUS and methane reduction in enabling near-term decarbonization and advancing the industry's net-zero ambitions.
- vi. **Operational Integration of Sustainability** module demonstrates how sustainability is embedded into daily operations through real-world examples, linking tools, targets, and decision-making to measurable environmental outcomes.
- vii. **Global Market and Regulatory Outlook** module provides insight into evolving regulatory frameworks and market expectations shaping the future of sustainable energy delivery.
- viii. **Innovation and Sustainability - Driven Thinking** module encourages students to apply their technical learning toward proposing innovative solutions aligned with sustainability priorities.

Carbon Capture and Underground Storage – Interpretation and Modeling (Screening and Monitoring)

Building on the knowledge and skills acquired in previous training sessions, learners will apply their expertise to a real-world case study involving subsurface CO₂ storage. They will undertake geological characterization, develop modeling workflows, and conduct both rapid and detailed screening of the site. The exercise will culminate in implementing monitoring qualitative techniques to evaluate plume behavior and ensure long-term storage integrity.

This module explores the key principles of CO₂ movement and containment in the subsurface, with a strong emphasis on modeling techniques, screening methodologies and monitoring tools essential to Carbon Capture, Utilization and Storage (CCUS/CCS) applications.

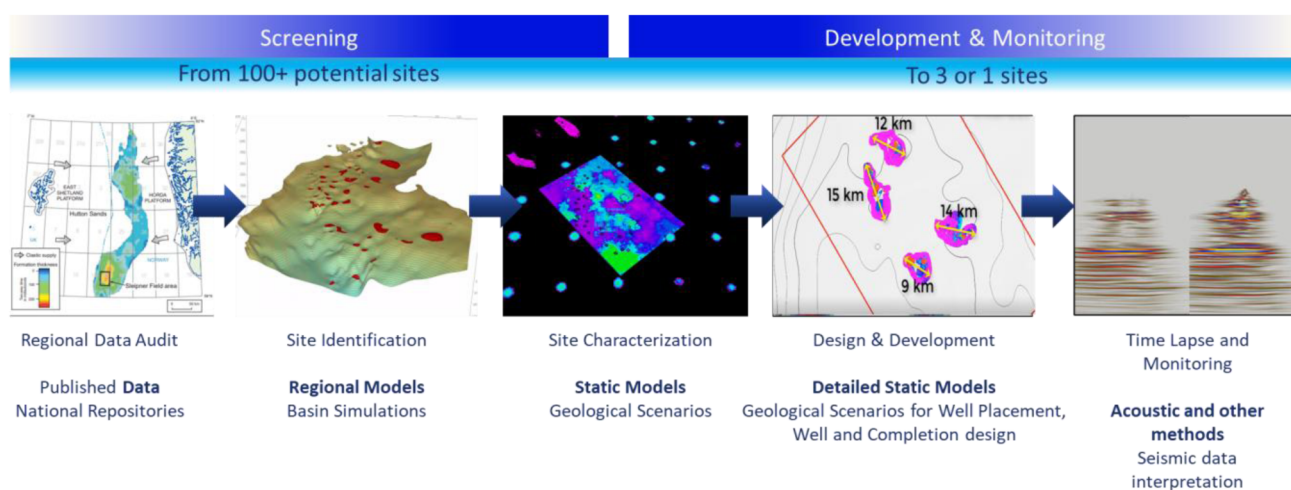


Figure 5—Geosciences path for a successful storage project

Detailed modules as below:

- i. **Introduction to Basin Modeling** module begins with an overview of industry-standard basin modeling tools used to simulate geological evolution and fluid dynamics. Learn their applications in CO₂ migration, trapping mechanisms and risk analysis.
- ii. **Applications in CCUS/CCS Projects** module covers how subsurface modeling supports carbon storage initiatives, including site characterization, containment evaluation, and regulatory compliance.
- iii. **Screening and Monitoring Principles** module explores theoretical approaches for identifying and assessing potential CO₂ storage formations, including criteria for screening, monitoring technologies, and regulatory frameworks.
- iv. **Plume Prediction Techniques (Rapid Screening)** module covers preliminary assessments of plume dispersion using fast and simplified models. Gain insight into plume geometry, movement, and containment efficiency for early-stage project evaluation.
- v. **Detailed 3D Geological Modeling** module showcase high-resolution 3D models to simulate complex geological scenarios. Focus on lithological setup, reservoir parameters, and fine-tuned predictions of plume behavior under various injection conditions.
- vi. **Workflow Integration with Reservoir Characterization Tools** module connects simulation results with broader reservoir modeling platforms to ensure continuity in project planning, subsurface visualization, and performance analysis.
- vii. **Time-Lapse Monitoring Strategies** module covers how temporal subsurface data—such as 4D seismic (borehole and surface)—can be used to monitor CO₂ movement and ensure integrity over time.

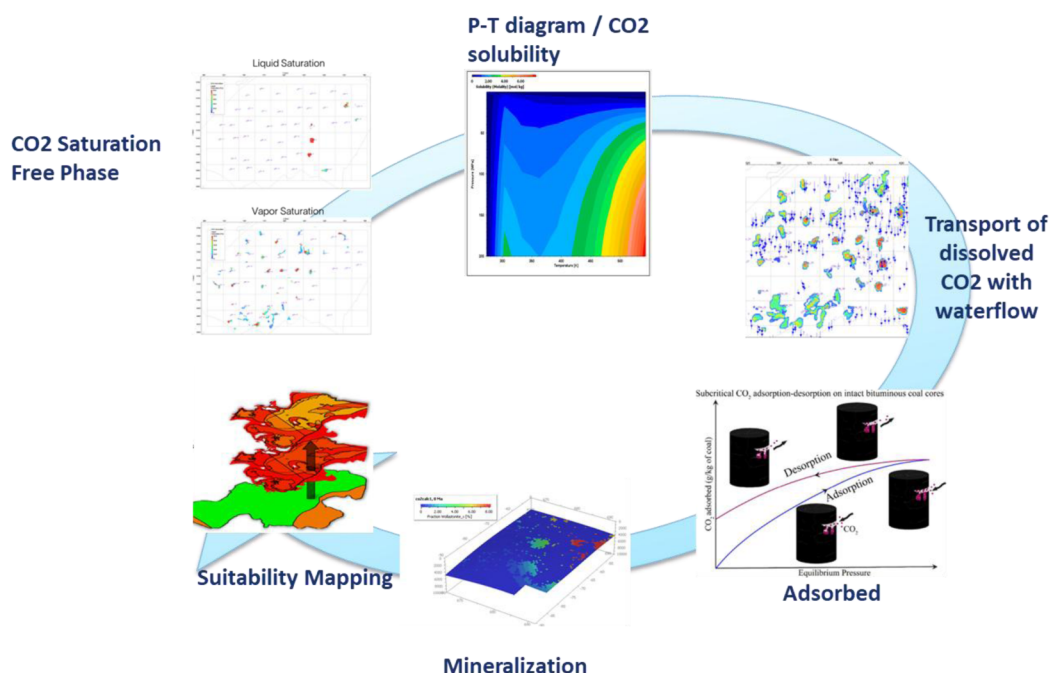


Figure 6—Capacity assessment – Trapping Mechanisms workflow.

CO₂ Modeling and Interpretation equips geoscientists with the analytical framework and technical precision needed for subsurface carbon storage. Through modeling, screening and time-lapse monitoring, professionals gain the ability to predict and track plume dynamics, assess reservoir integrity, and ensure long-term storage security. This knowledge doesn't just support climate resilience; it places geoscientists at the forefront of designing effective, scalable solutions for a low-carbon future.

Log-based CO₂ Monitoring with Pulsed Neutron Measurement

Reservoir monitoring through Pulsed Neutron Log (PNL) in case hole environments has been an integral part of petrophysics trainees learning journey. This module is now expanded to cover CCS applications for monitoring the changes in CO₂ fluid saturation in the pore space over time. The module is focused on showcasing a technique that observes the changes reflected on multiple measurements, namely the thermal neutron capture cross section (Sigma), fast neutron cross section (FNXS), and the thermal neutron porosity (TPHI), due to CO₂ injection, as described by Laronga et al (2023) in Figure 7.

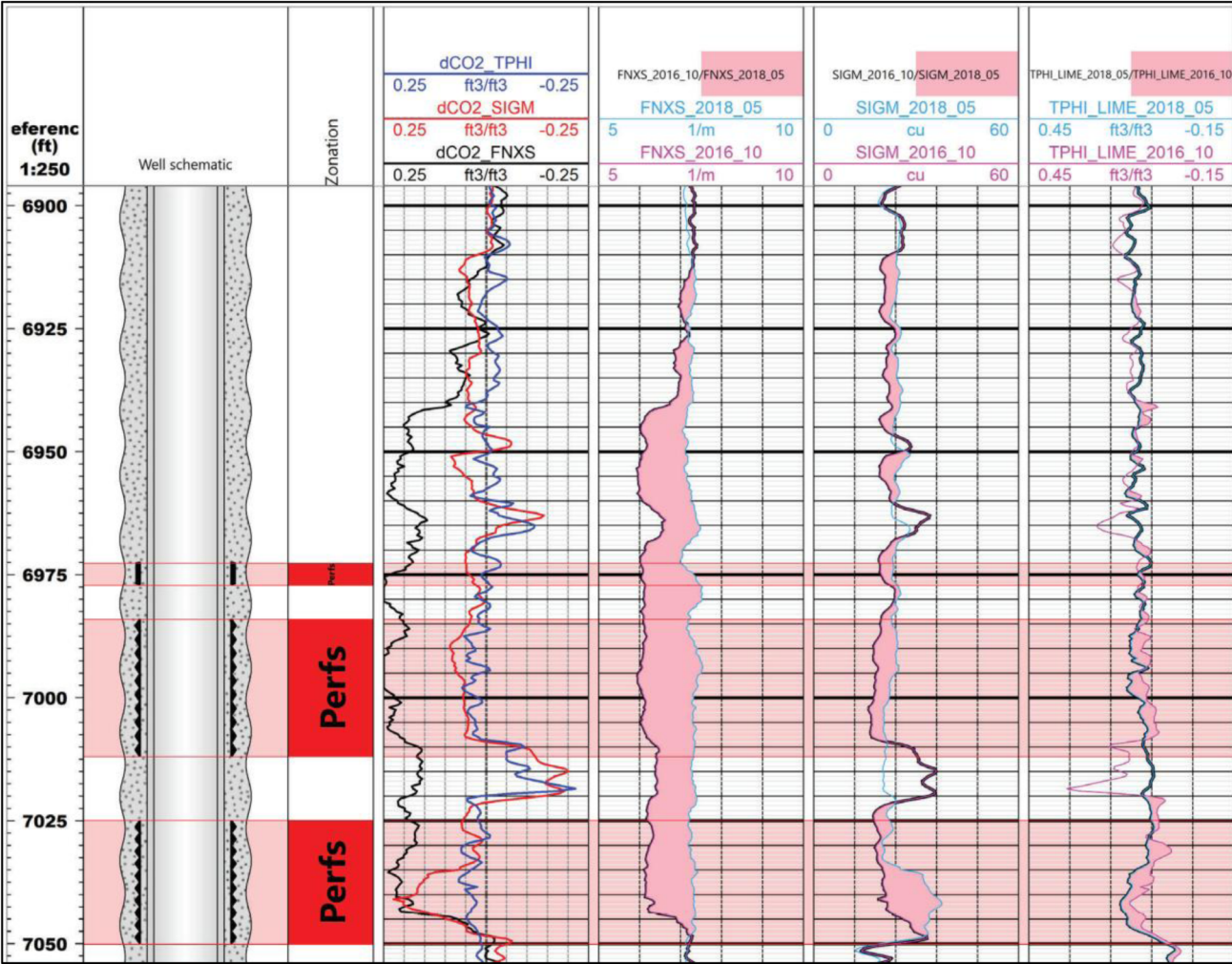


Figure 7—Observing changes in Pulsed Neutron Log measurement (FNXS, Sigma, and TPHI) over the span of two years. The Increment in CO₂ pore volume is denoted by the pink shading against and around the perforation zones (Laronga et al 2023)

The learning module also covers the quantification of the CO₂ and native reservoir fluids distribution at different time steps through multi-staged simultaneous inversions that integrate a multitude of standard and advanced open hole and cased hole log measurements, following the scheme described in Figure 8.

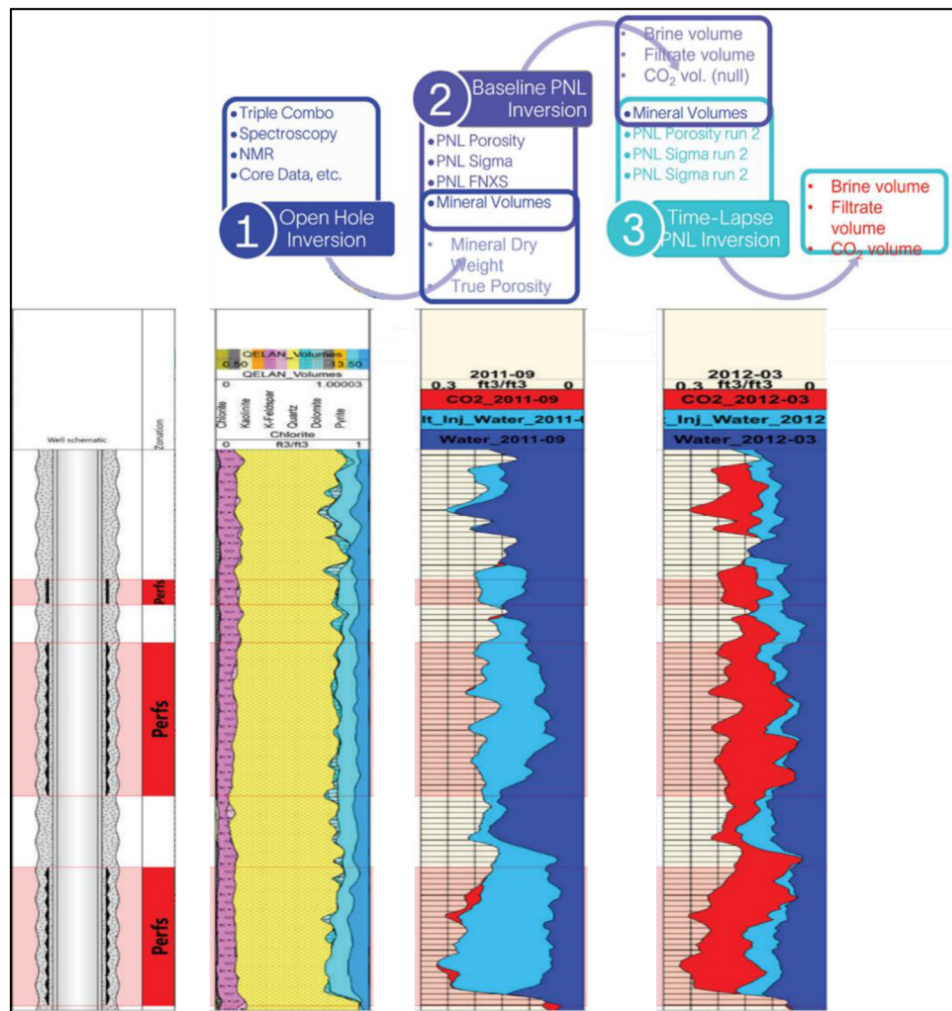


Figure 8—Multi-staged inversions of open hole and cased hole logs to provide volumetric quantification of pore fluids prior and post CO₂ injection (CO₂ denoted by red shading in last track, modified from Laronga et al, 2023).

Inclusion of this CCS focused log-based learning module is critical for petrophysics trainees to bridge pertinent competency gaps against the rising business demand for PNL CO₂ monitoring.

Coupled Modeling for Subsurface Storage Simulation

This advanced module builds upon the foundation of traditional subsurface reservoir modeling, expanding its scope to support emerging applications in carbon and hydrogen storage. The training journey begins with established industry practices, including geomodelling and rock–fluid physics, before progressing toward more specialized applications.

Participants first revisit core concepts fundamental to the oil and gas industry, such as advanced Pressure-Volume-Temperature (PVT) analysis. This includes detailed instructions on fluid characterization using equations of state and flash modeling techniques. These skills are essential for understanding multiphase behavior in subsurface environments that build the foundation towards understanding carbon dioxide and hydrogen fluid behavior in the reservoirs.

The curriculum then transitions into the specialized domain of subsurface storage, where learners explore the scientific principles and simulation methodologies required to model carbon dioxide and hydrogen injection and containment. Central to this is a detailed treatment of key trapping mechanisms:

- i. Structural Trapping: Utilizing geological features such as anticlines to physically contain injected gases.

- ii. Residual Trapping: Relying on capillary forces to immobilize fluids within pore spaces.
- iii. Solubility Trapping: Enabling injected gases to dissolve into formation fluids over time.
- iv. Mineralization: Modeling the long-term geochemical reactions that convert CO₂ into stable carbonate minerals.

These mechanisms are explored through practical exercises using industry-standard subsurface modeling platforms that enable high fidelity and physics-based modeling with superior performance and no-coding requirements.

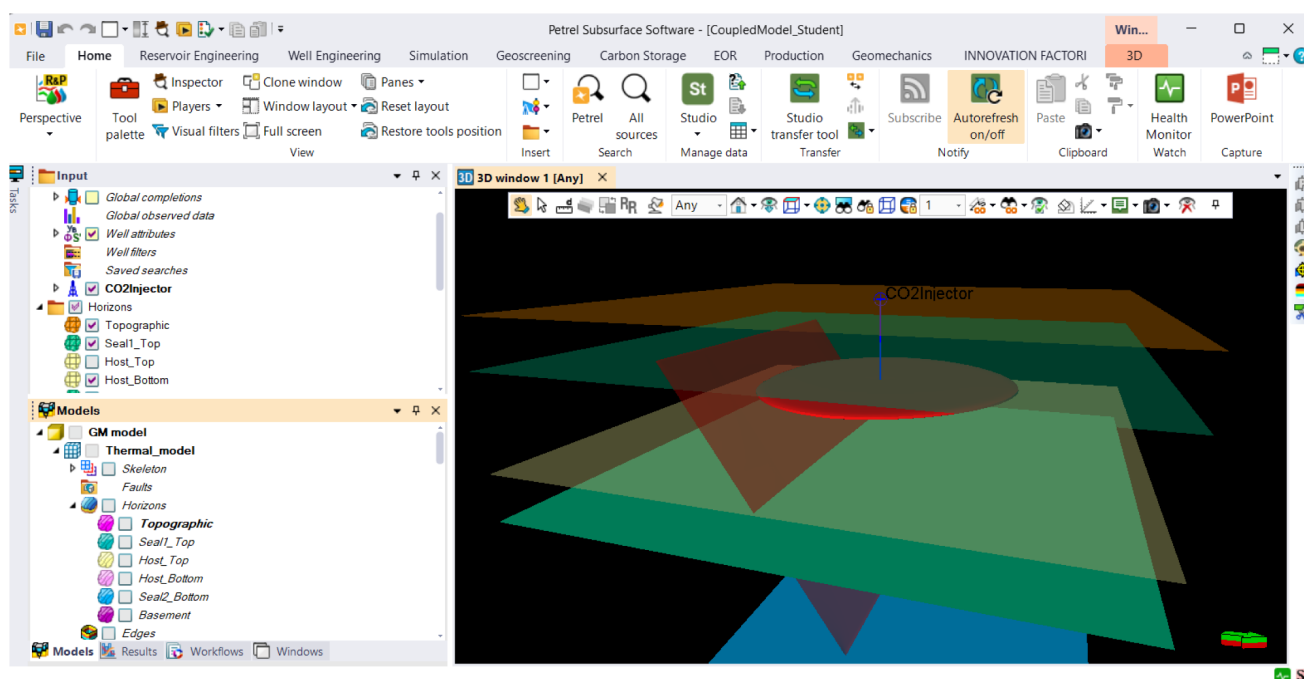


Figure 9—Coursework with synthetic data to model CO₂ plume migration in subsurface environment.

The curriculum then transitions to integrating geomechanical and reservoir simulation engines. Learners are trained to couple fluid flow simulations with geomechanical models to evaluate critical storage parameters, such as fault stability and caprock integrity, under various injection scenarios.

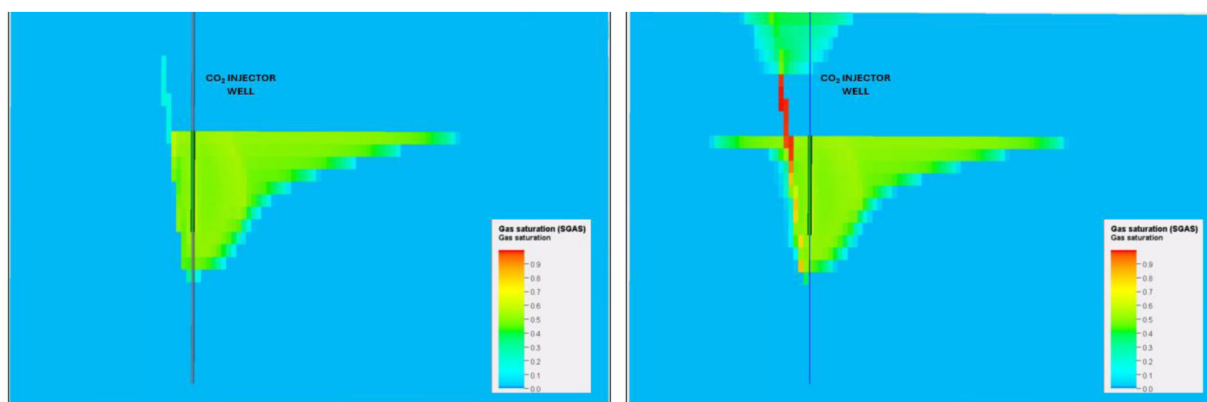


Figure 10—Coursework with synthetic data to study the impact of integrating thermal, hydro and mechanical aspects (on the right) as opposed to without integration (on the left) for carbon storage scenarios.

This integrated approach ensures participants are equipped with the tools and insights necessary to design safe, efficient, and regulatory-compliant storage projects. By bridging traditional petroleum engineering techniques with modern subsurface storage applications, the module fosters cross-disciplinary competencies essential for advancing carbon management and energy transition strategies.

Carbon Capture and Transport

This modular course focuses on building upon learner's prior knowledge on thermodynamics, phase behavior and flow assurance applied to oil and gas; to understand CO₂ behavior during its pipeline transport and the evaluation of different CO₂ capture techniques applying specialized software technologies through hands-on examples.

The evaluation for CO₂ transport and injection course is delivered by using a synthetic dataset evaluating the behavior of pure CO₂ and with impurities, under different conditions of pressure and temperature to design and analyze pipeline transport systems and injection at well level, investigating for flow assurance challenges, impact of operational changes and safety considerations. The course aims to apply the latest release of the dynamic multiphase flow simulator, including the new solver that can capture two-phase CO₂ flow more accurately, as well as the incorporation of industry best practices and guidelines.

CO₂ Capture, dehydration and compression operations have also been incorporated as part of the curriculum to provide the learners with hands-on examples applying process simulation as a tool to evaluate CO₂ capture alternatives, modeling dehydration processes and the compression strategy to handle CO₂ flow. The learner will be able to understand the impact of CO₂ sources and the capture technologies available; the dehydration needs to prevent any flow assurance challenges and even assess the impact of fluid flow properties (i.e. density) and presence of impurities in compression equipment requirements (i.e. intercooling requirements, power consumption) and safety hazards.

The incorporation of these modules into the curriculum has allowed increased proficiency and knowledge of production engineers focused on oil and gas operations, to contribute to projects related to CCS and global energy transition ambitions.

Carbon Storage Ranking and Screening as part of Digital Solutions for Energy Transition

This advanced module builds upon the foundational energy transition curriculum by exposing learners to real-world challenges through public data and hands-on application of cutting-edge digital technologies. A key component of the coursework is the use of a digital sustainability platform to perform carbon storage site screening and ranking.

The curriculum incorporates both synthetic and publicly available datasets on prospective carbon storage sites and major point-source emitters. Learners are guided through practical exercises that enable them to visualize spatial relationships, evaluate site characteristics, and simulate real-world workflows for carbon storage site screening and ranking.

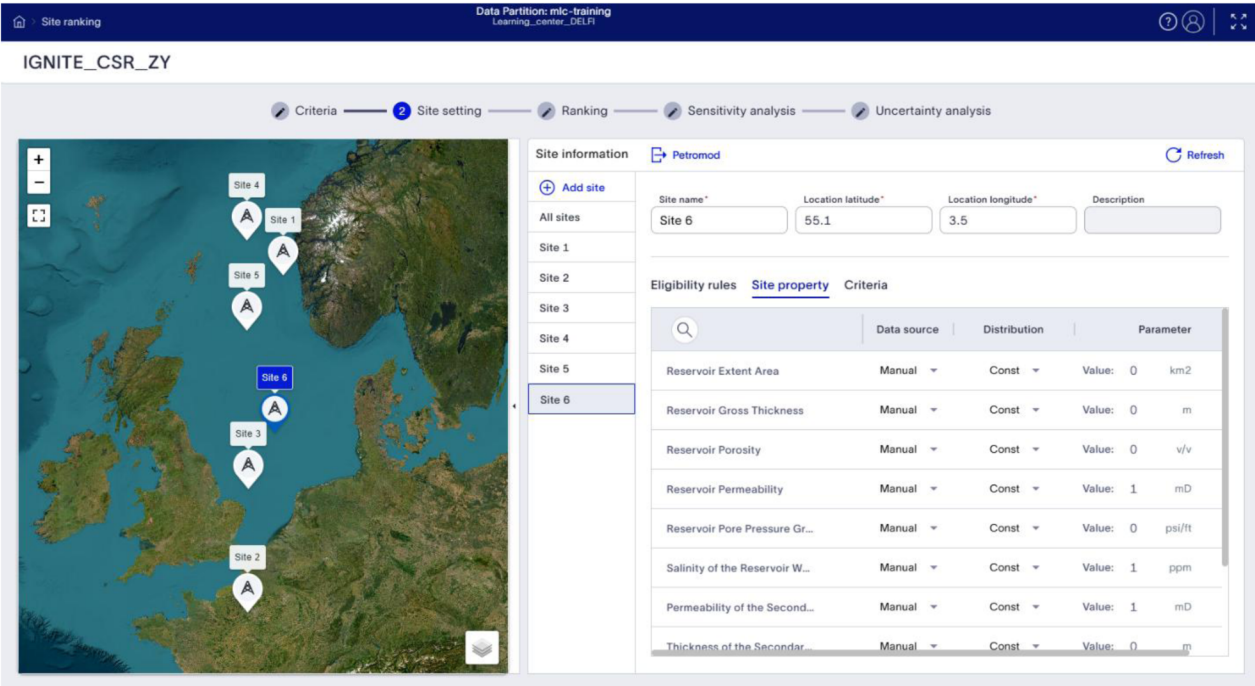


Figure 11—Coursework using digital sustainability platform for carbon storage site screening and ranking.

Participants apply structured decision-making methodologies such as the Analytic Hierarchy Process (AHP) and the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) in this module. These multi-criteria decision analysis (MCDA) tools help quantify trade-offs between competing factors and enhance the accuracy and objectivity of site selection processes (Ma, 2024).

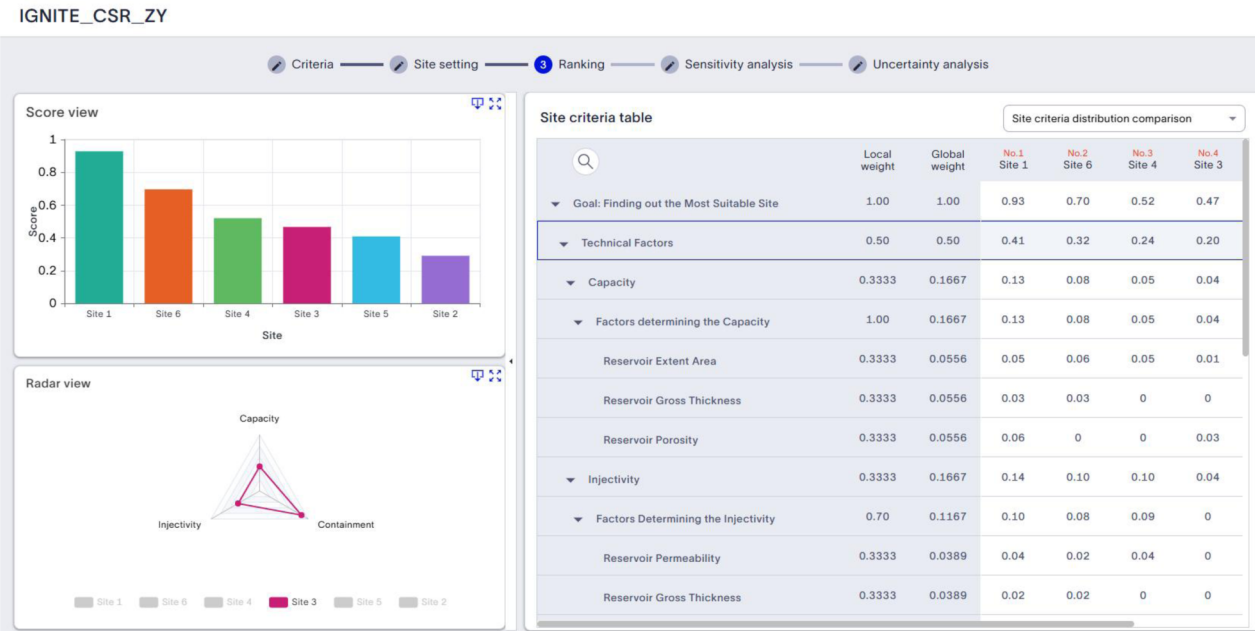


Figure 11a—Digital sustainability platform that enables carbon storage site screening and ranking.

To deepen analytical rigor, learners conduct sensitivity and uncertainty analyses using Monte Carlo simulation techniques. These analyses allow participants to evaluate how uncertainties in subsurface and infrastructure parameters influence the final ranking of candidate sites. Sensitivity analysis identifies the

most influential variables, guiding targeted data acquisition to reduce uncertainty and improve decision confidence (Ma. 2024).

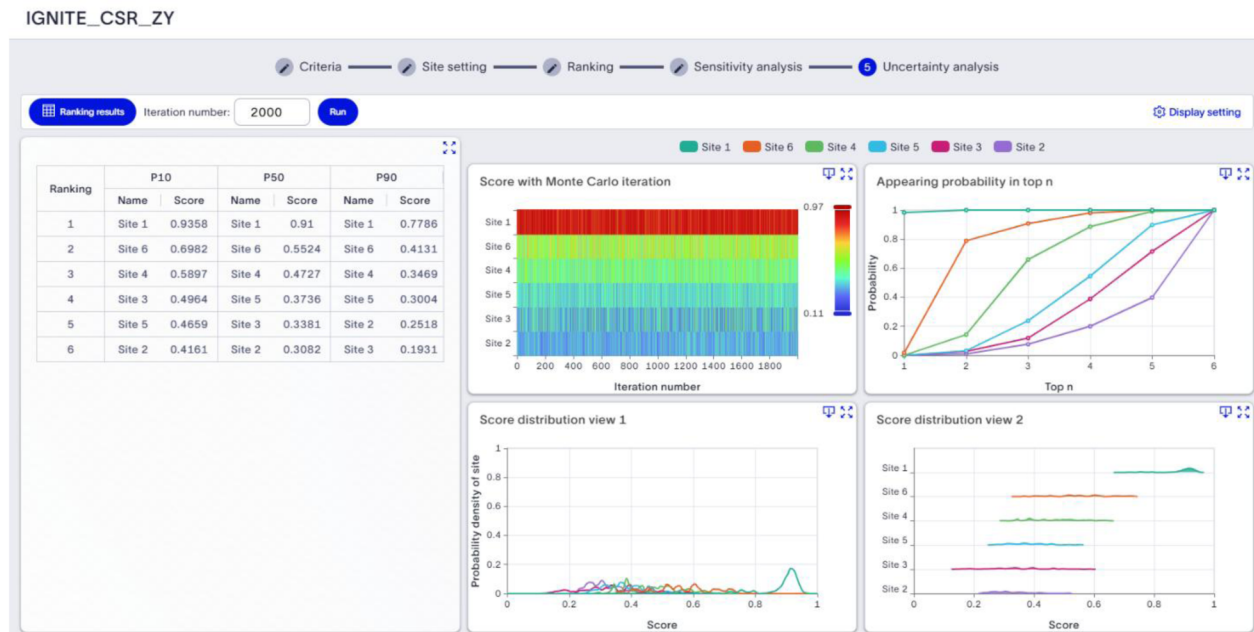


Figure 12—Learners perform sensitivity and uncertainty analysis to evaluate impact of various parameters affecting site score.

Through this module, learners gain practical experience in applying digital tools and methodologies that are essential for supporting high-quality, data-driven decisions in carbon storage planning. The integration of advanced analytics within a structured evaluation framework equips participants with the competencies required to accelerate low-carbon project deployment with greater precision and reduced risk.

Offshore Wind

Offshore wind energy is entering a golden age of growth, and traditional oil and gas geoscientists are uniquely positioned to help drive this transformation. Instead of hunting for hydrocarbons, they are now applying their deep subsurface expertise to understand seabed dynamics, guide foundation design and identify-mitigate geohazards. All essential tasks for the safe and efficient deployment of offshore wind farms.

In the coming years, global offshore wind capacity is set to grow over 400 GW by 2035. Regions like China, Northern Europe and parts of Asia-Pacific are leading the charge, unlocking enormous demand for skilled professionals who know their way around seismic data, geotechnical assessments and marine environments. Floating wind farms are also on the rise and here, geoscientists with deepwater O&G experience are especially valuable in anchoring turbines in challenging conditions.

To thrive in offshore wind energy, especially for geoscientists transitioning from oil and gas, learners must become fluent in interpreting specialized datasets central to site characterization. This includes ultra-high resolution (UHR) seismic data for shallow subsurface imaging and cone penetration test (CPT) logs for direct soil property assessment—both essential for safe turbine sitting and foundation design.

Other data types include:

- Bathymetric surveys for mapping seabed topography.
- Side-scan sonar and magnetometry for detecting geohazards (seafloor, shallow subsurface and synthetic).

- Geotechnical borehole data as cores for validating seismic interpretations and calibrating ground models.

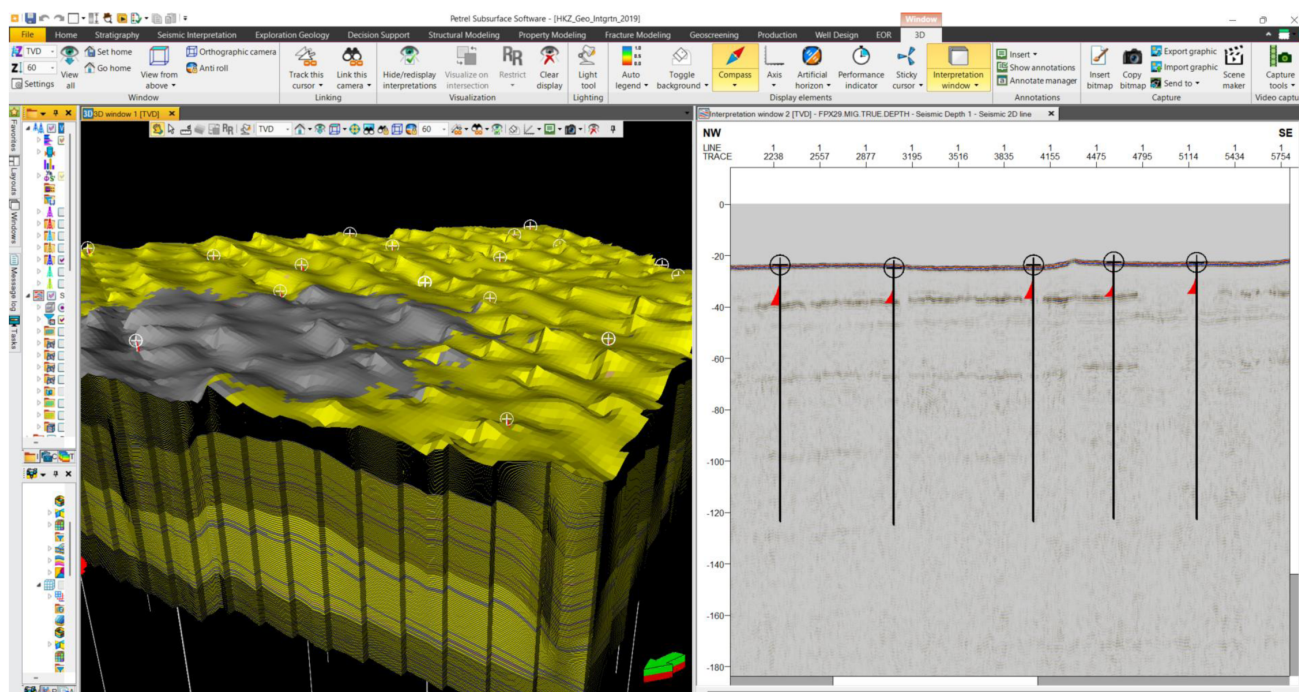


Figure 13—Learners engage with challenges analogous to those encountered in the Oil & Gas sector, but at a higher spatial resolution and with additional complexities unique to offshore wind energy systems.

These inputs are incorporated into a geological model, which undergoes structural and property modeling to capture subsurface features and spatial distributions. The resulting model is then passed on to geomechanics engineers, who use it to evaluate mechanical behavior, foundation performance and long-term site stability—essential for designing safe and resilient offshore infrastructure.

Emissions Management with Digital Solutions

Global energy-related CO₂ emissions reached a record 37.8 Gt in 2024, with nearly 90% arising directly from fossil fuel combustion in coal, oil, and gas sectors (IEA, 2025). Given that the energy system accounts for approximately three-quarters of all greenhouse gas emissions, the oil and gas industry have an essential responsibility to implement effective reduction strategies. Digital transformation plays a critical role in this effort; the International Energy Agency highlights that AI-driven technologies, such as predictive maintenance and real-time optimization, significantly reduce emissions across the energy value chain. Additionally, recent research by McKinsey (2025) suggests digital tools could accelerate up to 47% of the decarbonization required to achieve the 1.5°C climate target by 2050. Thus, developing talent capable of leveraging digital solutions is fundamental to achieving net-zero ambitions.

To address this need, the curriculum integrates a practical emissions management exercise (Fig.10) using the organization's AI platform. Learners ingest and analyze sensor data streams from various facilities, including training centers, production systems, processing plants, and transportation networks. Using recognized emission-factor libraries, participants convert sensor data into accurate CO₂-equivalent emissions inventories. Subsequently, learners apply AI-driven time-series forecasting models to simulate future emissions scenarios based on different operational conditions and projected energy consumption. Through iterative modeling and analysis, participants identify optimal operational adjustments—such as load balancing between assets, scheduling energy-intensive processes, or refining flaring protocols—to minimize emissions while maintaining safety and efficiency standards (Fig. 11).

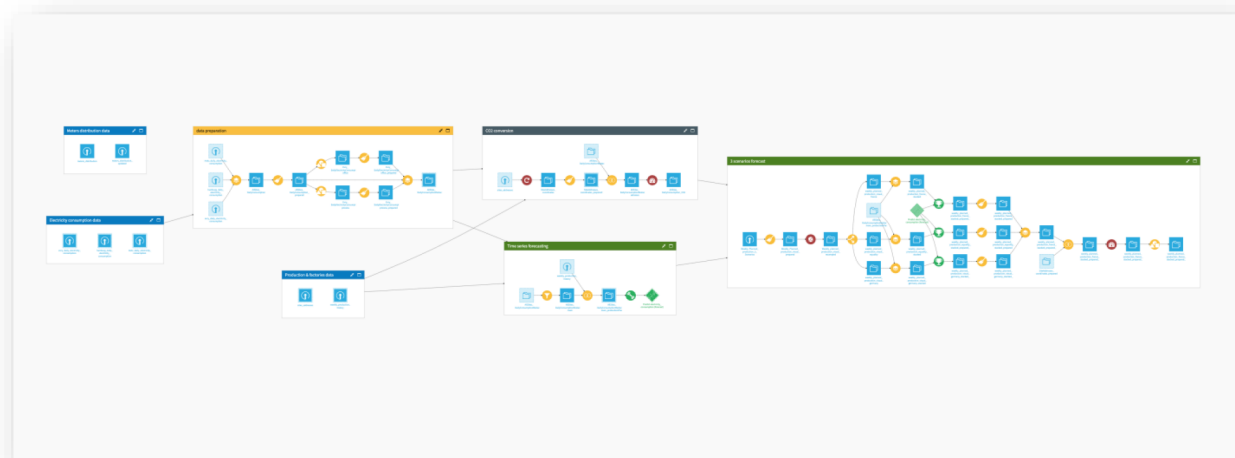


Figure 14—Learners model energy consumption to forecast electricity and CO2 emissions

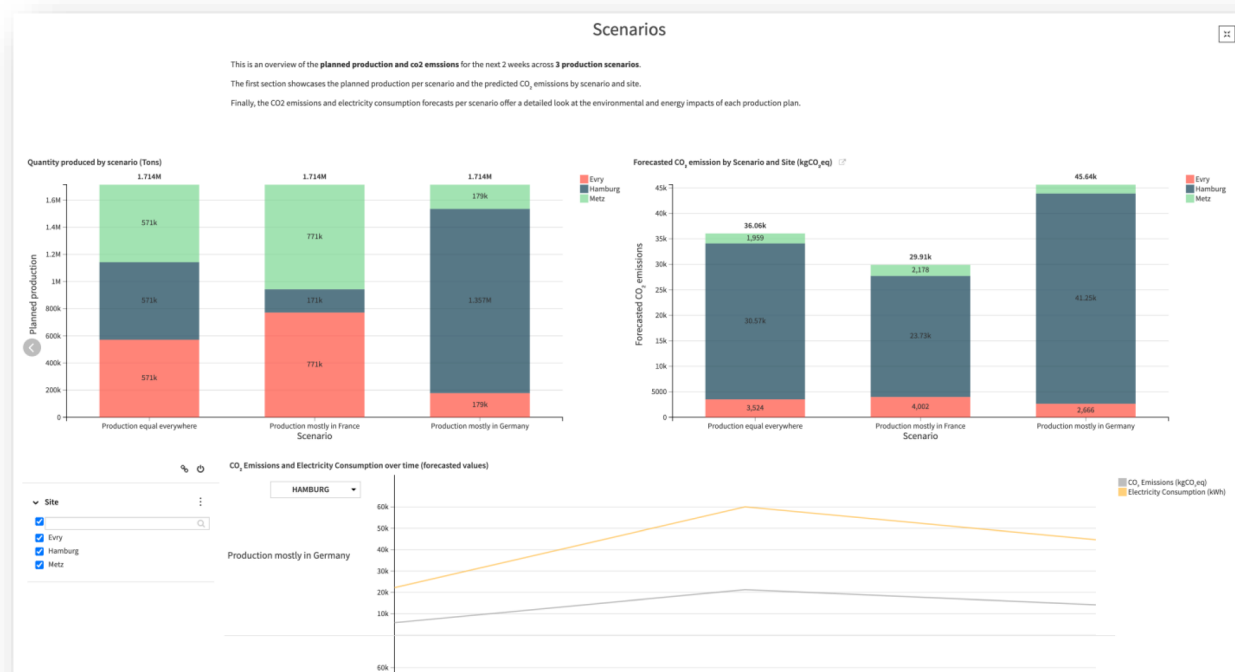


Figure 15—Learners conduct scenario-driven what-if analyses to identify operational adjustments that minimize CO₂-equivalent emissions under varying production plans and site configurations

Such practical, scenario-based training delivers significant benefits. First, learners acquire essential quantitative skills to convert operational data into verifiable emissions metrics, increasingly mandated by regulators and stakeholders. Second, the exercise promotes a systems-thinking approach, allowing participants to understand surface and subsurface assets as interconnected elements whose emissions performance must be optimized collectively rather than in isolation. Lastly, integrating AI analytics into routine decision-making equips the workforce to lead innovative, cross-disciplinary initiatives effectively, driving organizational progress toward net-zero commitments and supporting broader sustainability objectives.

Conclusion

The transformation of the petrotechnical training programme demonstrates that the shift toward low-carbon energy systems is not about discarding legacy knowledge, but rather about evolving it. By reconfiguring workflows, emphasizing transferable physics-based principles, and integrating digital technologies, organizations can prepare their workforce to tackle the energy transition with both confidence and competence.

The revamped curriculum highlights the importance of domain integration and modular learning, equipping professionals with interdisciplinary skills that transcend traditional oil and gas silos. Practical immersion—through digital field trips, lab experiences, and advanced tool interactions—ensures that learning remains grounded in real-world application.

This paper has outlined how our organization has embraced this challenge by reshaping corporate talent development around current and future business needs. The outcome is a scalable, adaptable framework capable of nurturing future-ready professionals who can deliver across traditional and emerging energy projects alike.

As industry continues to navigate an increasingly complex and integrated energy landscape, one thing is clear: new energies will require new technologies, new methods, and a deeper level of cross-domain integration. The transition is technical and a transformation of the mindset. It demands new skills, reimagined workflows, and the deliberate development of the next generation of energy professionals equipped to lead this change.

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